

Sheikh Rahat Mahmud

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[sheikh-rahata-mahmud](#) — 🌐 [Portfolio](#)

Education

Bangladesh University of Engineering & Technology (BUET), Dhaka **Expected June 2026**
Bachelor of Science in Computer Science and Engineering **Final Year (L-4 T-2)**
CGPA: 3.11 / 4.00 (Last Term(L-4T-1) **GPA: 3.32 / 4.00**)

Higher Secondary Certificate (HSC) **GPA: 5.00 / 5.00**

Secondary School Certificate (SSC) **GPA: 5.00 / 5.00**

Relevant Coursework:

- **Core Systems:** Computer Networks, Info. System Design, Software Engineering, Microcontrollers & Embedded Systems, Computer Architecture, Operating Systems.
- **Mathematical & Analytics:** Statistics, Discrete Math, Calculus, Linear Algebra, Economics.
- **Industrial & Business:** IPE (Industrial & Production Engineering) course, Accounting & Finance, Electronics, Electrical Machines, English course.

Technical Skills

- **Languages:** Python, JavaScript/TypeScript, Java, C++, SQL
- **Frontend:** React.js, React Native, Material UI, HTML/CSS
- **Backend:** FastAPI, Django, Express.js, REST APIs, Payment Gateway Integration
- **Databases:** PostgreSQL, Oracle, MongoDB
- **DevOps & Tools:** Git, CI/CD Pipelines, Docker, Linux, VM Deployment
- **AI Frameworks:** LangChain, LangGraph
- **ML/DL:** PyTorch, CNN, Neural Networks, Transformer, Knowledge Distillation
- **Explored & Used Models:** ResNet, UNet, 3D-UNet, Swin-UNet, UNETR, DNABERT-2, YOLOv8, YOLOv11, various ASR Models.
- **Other:** Agile Development, Code Review

Experience

Software Development Intern **Present**
CalHVAC, New York (Remote)

- Developing full-stack solutions using Python backend , modern JavaScript framework for frontend & Three.js for simulation.

Open Source Developer **Sept 2025 – Present**
Collaborating with Senior Software Engineer from renowned company (Remote)

- Contributing to React Native mobile and Electron macOS desktop applications.
- Implemented WatermelonDB for offline-first data persistence in mobile apps.

Intern **Nov 2025 – Dec 2025**
Bangladesh Satellite Company Limited (BSCL), Dhaka

- Built Python automation scripts for database retrieval, data processing, and visualization.

Payment Gateway Integration

- Integrated a secure payment gateway in a Node.js (Express.js) backend supporting transaction processing and verification.
- Implemented webhook handling and APIs for payment confirmation, order validation, and transaction status tracking.

Agentic AI Implementation

- Developed agentic AI workflows using LangChain and LangGraph enabling multi-step reasoning and tool-based task execution.
- Implemented modular LLM agents with memory and tool-calling for automated data processing and decision workflows.

Software Development Intern

Completed

Academic & Admission Pathshala (EdTech), Dhaka

- Improved front-end performance through code optimization and component refactoring.

Projects

Thesis: Auto Annotation of 3D QCT Image Using Deep Learning

Present

Supervisor: Dr. Mahmuda Naznin, Professor, CSE, BUET

- Developing deep learning models for automated 3D medical image annotation using architectures such as 3D-UNet, UNet, Swin-UNet, UNETR etc.

Lightweight Knowledge Distillation of Transformer Models for Biological Sequence Classification

Ongoing

Machine Learning Research Project

- Applying knowledge distillation techniques to compress large transformer models for efficient biological sequence classification.
- Working with DNABERT-2 and transformer-based architectures to achieve lightweight inference without significant accuracy loss.

IEEE Video & Image Processing (VIP) Cup

Participant

Competition

- Explored and applied various deep learning architectures including YOLOv8, YOLOv11, TransNetV2 for video and image processing tasks.

Career Studio– *React.js, FastAPI, CI/CD, VM Deployment*

[GitHub](#)

- Architected AI-powered CV and portfolio builder with FastAPI backend and React.js frontend.
- Implemented CI/CD pipeline for automated deployment to VM with AI content suggestions.

FaceBuet– *React.js, Django, Oracle, Material UI*

- Developed full-stack social media platform for BUET community with modular React.js components.
- Engineered Django REST backend with Oracle database and optimized SQL queries.

MovieDatabase– *Java, JavaFX, OOP Design Patterns*

- Built a desktop application with search, filter, and analytics features using object-oriented design principles.

Certifications

- [CUET Datathon 2025 – Finalist](#)
- [BSCL Internship Completion Certificate](#)
- [Solvio AI Hackathon – Round 2 Qualifier](#)
- [CUET AI Hackathon 2025 – Participant](#) (Explored various ASR models for speech recognition tasks)